

EPL Pick'em

Software Design & Development Document (v1.0)

1. Project Overview

Objective

Build a web application that replaces the current WhatsApp + Excel workflow used for our private Premier League Pick'em game.

The application should:

- Allow players to submit picks privately.
- Hide picks until reveal.
- Automatically calculate scores.
- Maintain league standings.
- Support Prediction Boosts and Late Passes.
- Provide complete auditability for all admin actions.
- Maintain an event history of the league.

This is intended for a **private group of friends (3–10 players)** and is optimized for simplicity rather than large-scale usage.

2. Tech Stack

Frontend

- React
- TypeScript
- Vite
- TailwindCSS
- shadcn/ui
- TanStack Query
- React Hook Form
- Zod

Backend

- NestJS
- TypeScript

Database

- PostgreSQL
- Prisma ORM

Authentication

- JWT
- Refresh Tokens
- bcrypt

Hosting

Frontend

- Vercel

Backend

- Render

Database

- Neon PostgreSQL

Repository

- GitHub
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3. Roles

Player

Can

- Login
 - Submit picks
 - Edit picks
 - Use Prediction Boosts
 - Use Late Passes
 - View standings
 - View history
 - View Event Log
-

Admin

Can

- Manage players
 - Promote/Demote Admins
 - Configure league
 - Manage fixtures
 - Import fixtures
 - Enter/Edit scores
 - Override picks
 - Recalculate standings
 - View Audit Logs
-

4. Game Rules

Season

- 20 EPL Teams
- 38 Gameweeks

Each player submits exactly one club every Gameweek.

Team Selection Rule

Players **cannot choose the same team in consecutive Gameweeks.**

Example

GW10 Arsenal

GW11 Arsenal ❌

GW12 Arsenal ✅

There are no other restrictions on club reuse.

5. Scoring

Match Result

Win = +3

Draw = +1

Loss = 0

Goal Difference Bonus

Win by 3+ goals = +1

Lose by 3+ goals = -1

Otherwise = 0

Examples

Liverpool 4-1

Base = 3

GD Bonus = +1

Total = 4

Chelsea 0-3

Base = 0

GD Bonus = -1

Total = -1

6. Prediction Boost

Every player receives

5 Prediction Boosts

per season.

When activated

Player must predict the exact score.

Example

Liverpool 3-1 Arsenal

If the prediction matches exactly

Weekly score is doubled.

Otherwise

Normal score applies.

Prediction Boost is consumed regardless of success.

7. Late Pass

Every player receives

3 Late Passes

Late Pass allows

- Submit after weekly deadline
- Edit after weekly deadline

Restriction

Player may only select a club whose fixture has **not kicked off**.

Once kickoff occurs

That fixture becomes unavailable.

No score penalty.

8. Weekly Lifecycle

Before Deadline

Players may

- Submit
- Edit
- Delete pick

Everyone else only sees

- Submitted
- Pending

Picks remain hidden.

Deadline Reached

Normal submissions lock.

Players with remaining Late Passes may continue editing/submitting.

Fixture Kickoff

Once a fixture starts

That club can no longer be selected.

Reveal

After the final fixture of the Gameweek kicks off

Reveal

- Club Pick
 - Prediction
 - Prediction Boost
 - Weekly Score
-

9. Dashboard

Displays

- Current Gameweek
 - Countdown
 - Deadline
 - Your Pick
 - Prediction Boosts Remaining
 - Late Passes Remaining
 - League Standings
 - Submission Status
 - Upcoming Fixtures
 - Recent Activity
-

10. Standings

Display

- Rank
 - Player
 - Total Points
 - Weekly Points
 - Aggregate Goal Difference
 - Prediction Boost Success Rate
 - Late Passes Remaining
-

11. Player Profile

Displays

- Season Points
 - Weekly History
 - Prediction Boost Usage
 - Late Pass Usage
 - Average Weekly Score
 - Highest Weekly Score
 - Total Wins
 - Draws
 - Losses
 - Aggregate Goal Difference
-

12. Weekly History

Every Gameweek shows

- Picked Club
 - Fixture
 - Final Score
 - Prediction
 - Prediction Result
 - Goal Difference
 - Weekly Points
 - Running Total
-

13. Submission Screen

Grid of EPL Clubs.

Each card displays

- Logo
- Opponent
- Home/Away
- Kickoff Time
- Availability

Bottom Section

- Prediction Boost Toggle
 - Score Prediction Inputs
 - Submit Button
-

14. Fixtures

Every fixture stores

- Home Team
- Away Team
- Kickoff
- Status
- Home Goals
- Away Goals
- Source

Source

- API
- Manual

Manual values always override API values.

15. Match Result Integration

Primary Source

football-data.org

Capabilities

- Fixtures
- Kickoff Times
- Scores
- Match Status

Fallback

16. Goal Difference Tracking

Each Pick stores

- Goals For
- Goals Against
- Goal Difference
- Match Result
- Base Points
- GD Bonus
- Multiplier
- Final Points

This preserves historical scoring even if rules change later.

17. League Settings

Configurable

- Season Name
 - Deadline Offset
 - Prediction Boost Count
 - Late Pass Count
 - Result Source
 - Reveal Mode
-

18. Admin Panel

Manage

- Users
 - Admins
 - Fixtures
 - Gameweeks
 - Results
 - Settings
 - Season
 - Score Recalculation
-

19. Admin Management

Admins may

- Promote Player
- Demote Admin
- Transfer Ownership

There must always be at least one Admin.

20. Event Log

Tracks every significant event.

Examples

- Player Submitted Pick
- Player Edited Pick
- Prediction Boost Used
- Late Pass Used
- Fixture Imported
- Fixture Updated
- Scores Calculated
- Reveal Completed
- Season Created

Event Log is visible to all players.

21. Audit Log

Tracks every administrative action.

Each entry stores

- Timestamp
- Admin
- Action
- Entity
- Previous Value
- New Value
- Reason (optional)

Examples

- Fixture edited
- Deadline changed

- Admin promoted
- Result overridden
- League settings changed

Audit Logs are immutable.

22. Database Model

User

- id
- name
- email
- passwordHash
- role
- createdAt
- updatedAt

Season

- id
- name
- year
- status

Team

- id
- name
- shortName
- logo

Gameweek

- id
- seasonId
- number
- deadline
- status

Fixture

- id
- gameweekId
- homeTeamId
- awayTeamId
- kickoff
- status
- homeGoals
- awayGoals

- source
- lastSyncedAt

Pick

- id
- playerId
- gameweekId
- teamId
- predictionBoostUsed
- predictedHomeGoals
- predictedAwayGoals
- latePassUsed
- submittedAt
- goalsFor
- goalsAgainst
- goalDifference
- matchResult
- basePoints
- gdBonus
- multiplier
- totalPoints

LeagueSettings

- id
- predictionBoostCount
- latePassCount
- deadlineOffset
- revealMode
- resultSource

EventLog

- id
- timestamp
- eventType
- userId
- gameweekId
- details(JSON)

AuditLog

- id
- timestamp
- adminId
- entity
- entityId
- action
- oldValue(JSON)
- newValue(JSON)
- reason

23. Backend Modules

- Auth
 - Users
 - Teams
 - Seasons
 - Gameweeks
 - Fixtures
 - Picks
 - Standings
 - Scoring
 - Events
 - Audit
 - League Settings
 - Admin
-

24. Core Services

Rather than placing business logic in controllers, the backend should use services.

PickValidationService

Responsible for

- Consecutive pick validation
 - Deadline validation
 - Late Pass validation
 - Fixture kickoff validation
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ScoringService

Responsible for

- Result points
 - Goal Difference bonus
 - Prediction Boost
 - Weekly totals
 - Standings
-

FixtureSyncService

Responsible for

- Importing fixtures

- Importing scores
 - API synchronization
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RevealService

Responsible for

- Pick visibility
 - Weekly reveal
-

EventService

Responsible for creating Event Log entries.

AuditService

Responsible for creating Audit Log entries.

25. API Endpoints (High Level)

Authentication

- POST /login
- POST /register
- GET /me

Teams

- GET /teams

Gameweeks

- GET /gameweeks

Fixtures

- GET /fixtures/:gameweek

Picks

- POST /pick
- PATCH /pick
- GET /pick/current

Standings

- GET /standings

History

- GET /history

Admin

- POST /fixtures/import
 - POST /results
 - PATCH /deadline
 - PATCH /settings
 - POST /recalculate
-

26. Frontend Pages

- Login
 - Register
 - Dashboard
 - Submit Pick
 - Standings
 - Player Profile
 - Weekly History
 - Admin
 - Settings
-

27. Validation Rules

- Cannot submit after deadline without Late Pass.
 - Cannot edit after selected team's kickoff.
 - Cannot select previous week's team.
 - Cannot exceed available Prediction Boosts.
 - Cannot exceed available Late Passes.
 - Manual scores override API scores.
 - Audit Logs cannot be deleted.
 - League must always have at least one Admin.
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28. Deployment

GitHub

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Vercel (Frontend)



Render (NestJS API)



Prisma ORM



Neon PostgreSQL

29. Development Roadmap

Sprint 1 – Foundation

- Repository setup
- Prisma schema
- Authentication
- User management

Sprint 2 – League Engine

- Teams
- Fixtures
- Gameweeks
- League settings

Sprint 3 – Gameplay

- Pick submission
- Validation
- Late Passes
- Prediction Boosts
- Event Log

Sprint 4 – Scoring

- Result integration
- Manual score entry
- Standings
- Goal Difference tracking
- Audit Log

Sprint 5 – Frontend

- Dashboard
- Submission screen
- Standings
- History
- Admin Panel

Sprint 6 – Deployment & Testing

- Football API integration
 - Deployment
 - End-to-end testing
 - Bug fixes
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30. Success Criteria

The application is complete when:

- Players can register and log in.
- Admin can create and configure a season.
- Fixtures are imported or entered manually.
- Players can submit and edit picks according to the rules.
- Picks remain hidden until reveal.
- Late Passes function correctly.
- Prediction Boosts correctly double scores.
- Goal Difference bonuses are applied automatically.
- Standings update automatically.
- Event Log records gameplay actions.
- Audit Log records administrative actions.
- The application is deployed and usable for the upcoming EPL season.